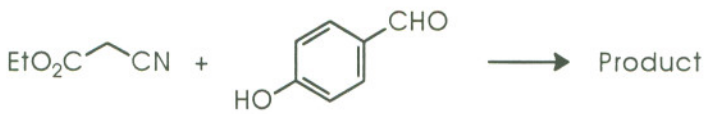
Organic Chemistry Cumulative Examination
January 20, 1996

This exam's topic is taken from Murahashi et al, JACS **1995**, *117*, 12436. The paper describes an extensive study of ruthenium-catalyzed aldol and Michael reactions of nitriles.

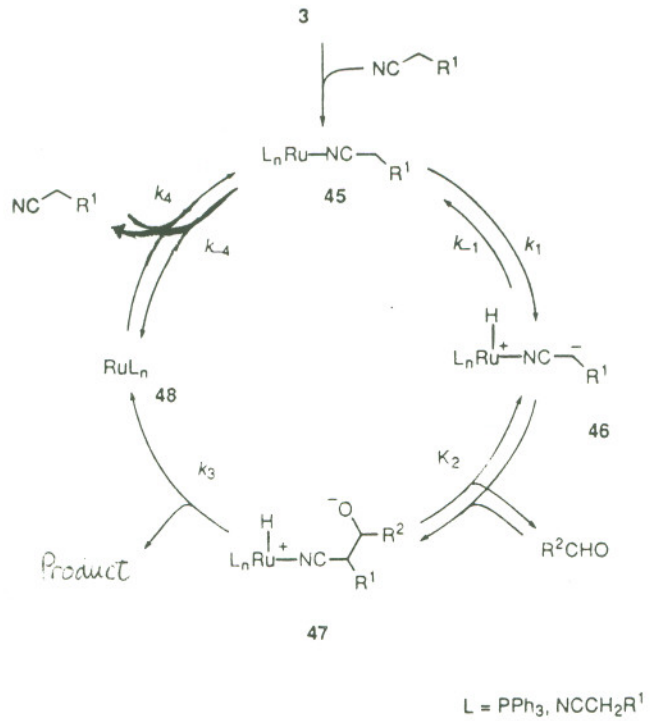
1. Draw the product(s) of the following aldol condensation reaction:



Draw the product(s) of the following Michael addition reaction:

2. Based on kinetic studies, the following mechanism for the catalyzed aldol condensation was proposed.

Scheme 1



The ruthenium catalyst is a phosphine complex, $RuH_2(PPh_3)_4$, shown as **3** in the mechanistic scheme above. The initial step involves coordination of the two nitrile molecules ($NCCH_2R^1$) to the ruthenium catalyst, replacing the hydrogens, forming the intermediate **45**. This is the active form of the catalyst; although it has two nitriles bound to the ruthenium, apparently only one undergoes the aldol reaction with the aldehyde (R^2CHO). When the product is formed, the catalyst is released in the form of a coordinatively unsaturated complex (**48**), which captures a new nitrile molecule to regenerate the active catalyst **45**.

The rate law is derived by assuming equilibrium for the oxidative addition to the α-CH bond of the nitrile (**45→46**), the addition to the aldehyde (**46→47**), and the coordination of the nitrile (**48→45**). In addition, it is also assumed that $K_2 << 1$; i.e., the intermediate **47** is never formed in significant quantities. The rate law so derived is:

$$\frac{d[\text{product}]}{dt} = \frac{k_3 K_2 [\text{Ru}]_{\text{total}} [\text{NCCH}_2\text{R}^1][\text{R}^2\text{CHO}]}{(k_{-1}/k_1 + 1)[\text{NCCH}_2\text{R}^1] + (k_{-1}/k_1)(k_{-4}/k_4)}$$

where $[\text{Ru}]_{\text{total}}$ is the total concentration of the catalyst in any of its forms (i.e., **45**, **46**, **47**, and **48**). Show the derivation of this rate law.

A further assumption is $k_{-1} << k_1$ and $k_{-4} << k_4$, which simplifies the rate law to one easily verified by experiment. Derive that simplified rate law.

3. Some experiments were performed to measure the rates of these ruthenium-catalyzed aldol reactions with para-substituted benzaldehydes. In the following table, those rates are listed for several substituents, along with the Hammett σ values for those substituents.

Substituent	log k	Hammett σ
H	−1.76	0.0
Me	−1.87	−0.17
Cl	−1.18	0.23
CN	−0.688	0.66

Sketch a rough Hammett plot of this data and determine, as best as you can, the ρ value for the ruthenium catalyzed reaction.